Creativity, Science and **Innovation**

Ideate, invent, innovate:
from ideas to prototypes

Additional material

Arduino Nicla SENSE ME

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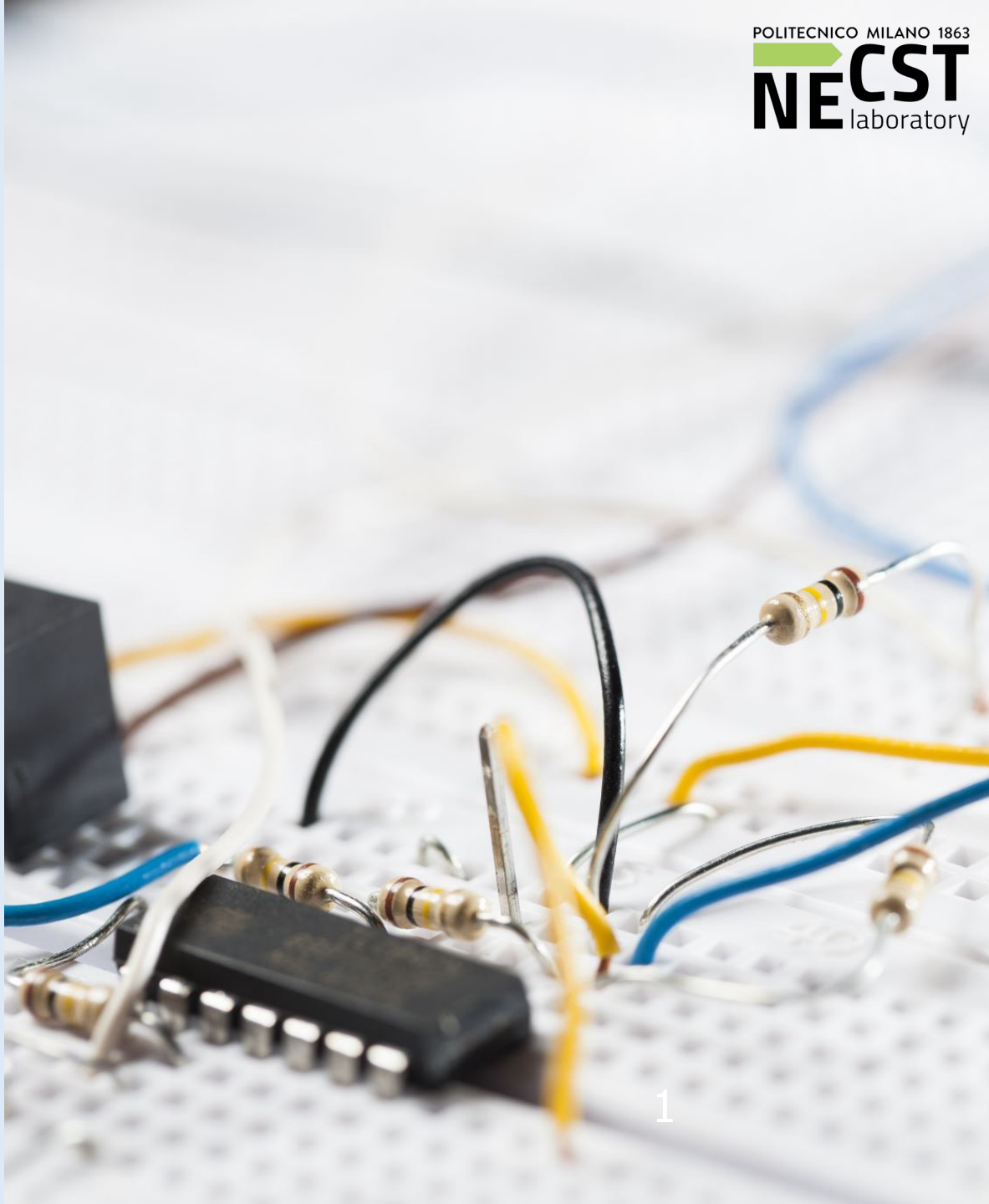
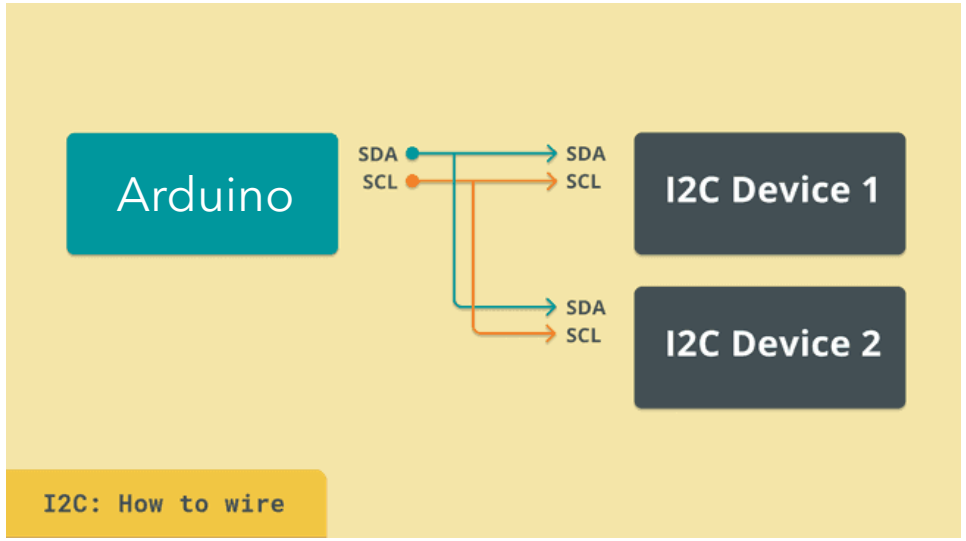


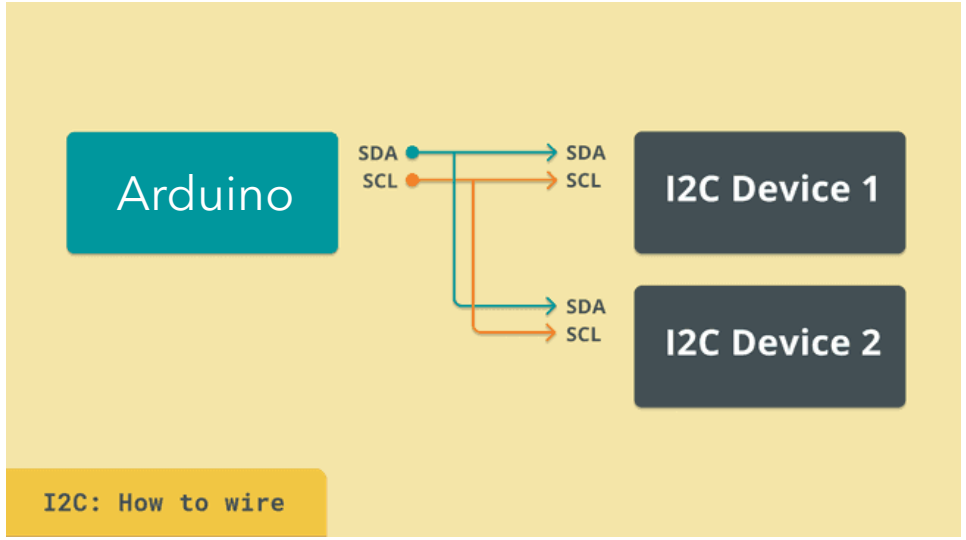
Inter-Integrated Circuit (I2C)



I2C is a serial protocol that allows you to connect many devices (up to 128) on the same two wires: Seria Data and Serial Clock.

Each component that uses I2C has a unique address of 7 or 10 bit).

Inter-Integrated Circuit (I2C)



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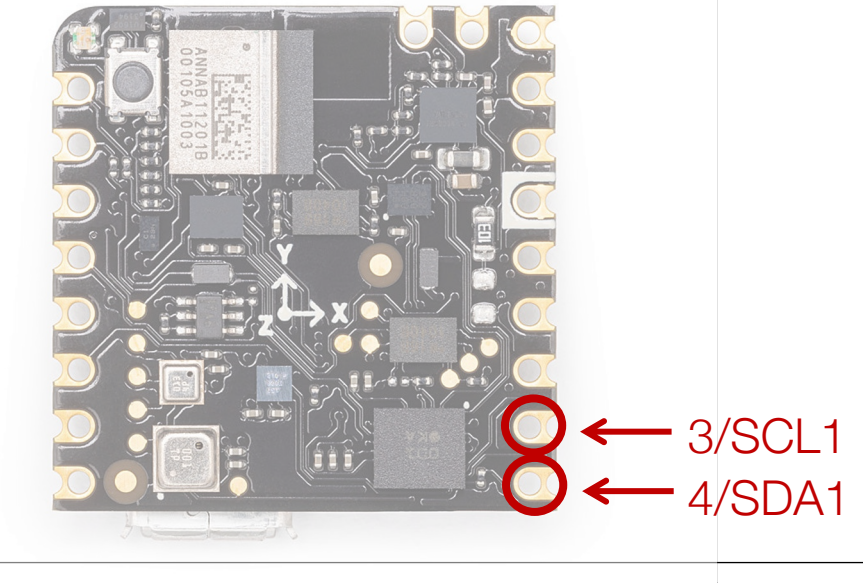
When the master communicates to a device:

- Start condition
- Device unique address (0X0C)
- Write/read flag (0/1)
- Data
- Stop condition

Often, data is accessed via internal registers inside the device, which have their own sub-addresses.



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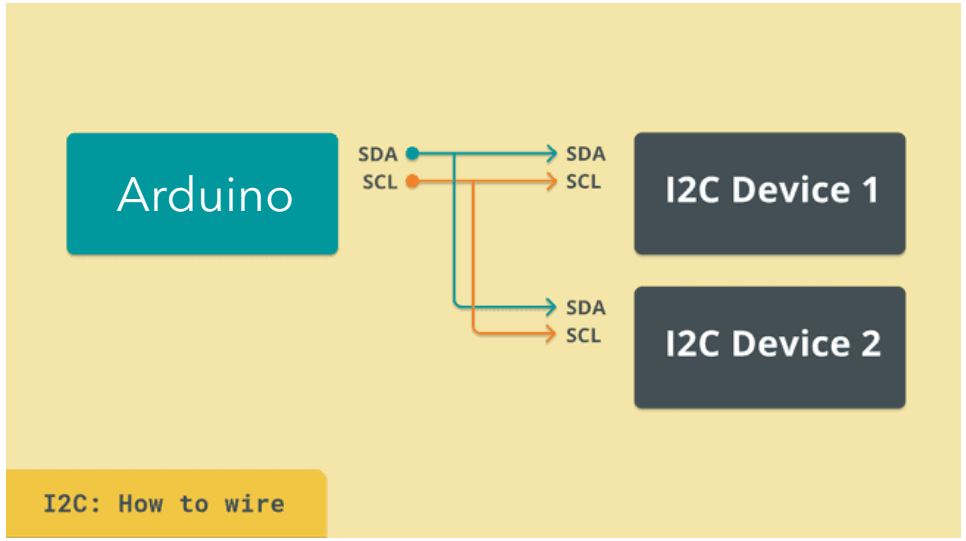

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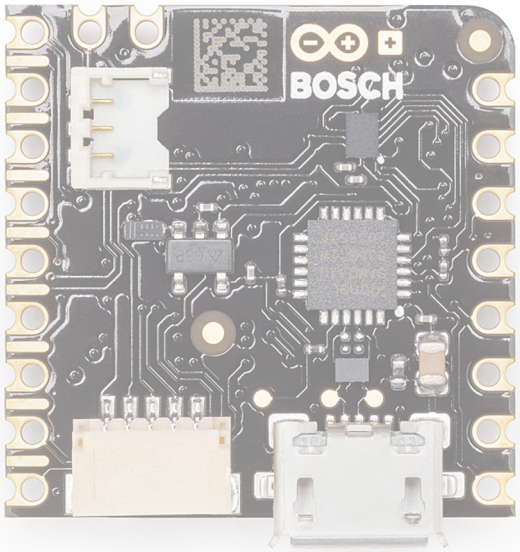


Inter-Integrated Circuit (I2C)

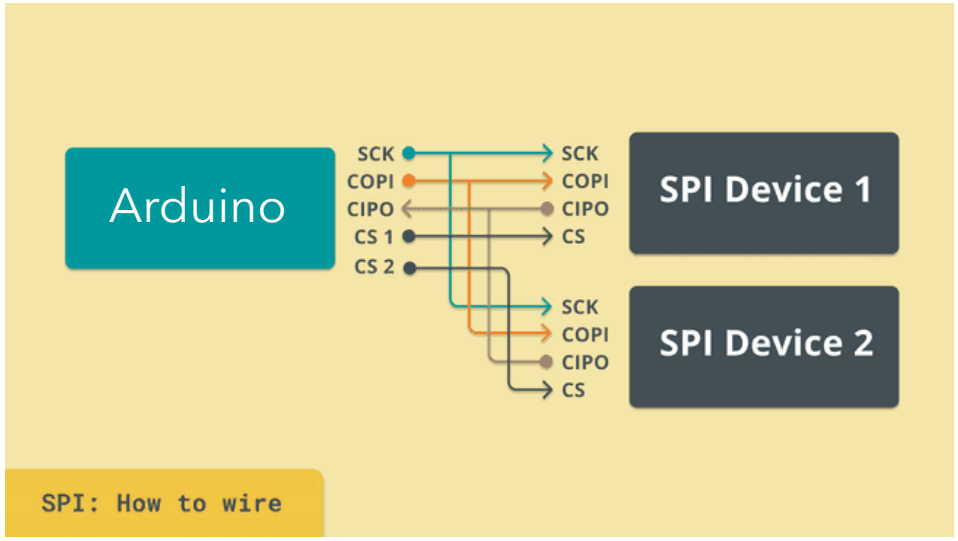
```
#include <Wire.h>

const byte deviceAddress = 0x35; // Target device's I2C address to which we want to communicate
const byte instruction = 0x00; // Instruction byte to be sent
const byte value = 0xFA; // Value to be sent
```

```
void setup() {
    Wire.begin(); // Initialize the SPI communication
}
```

```
void loop(){
    Wire.beginTransmission(deviceAddress); // Begin transmission to the target device
    Wire.write(instruction); // Send the instruction byte
    Wire.write(value); // Send the value
    Wire.endTransmission(); // End transmission
    }
```

Serial Peripheral Interface (SPI)



SPI is a 4-wire serial protocol and is also used for communication between systems.

It is in general twice as fast as I2C, but with more complexity in adding more devices to it.

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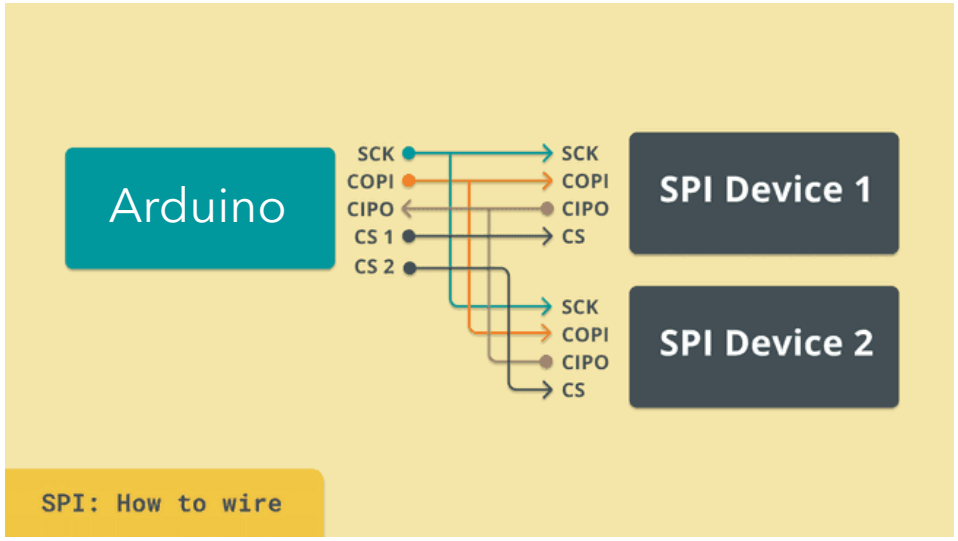

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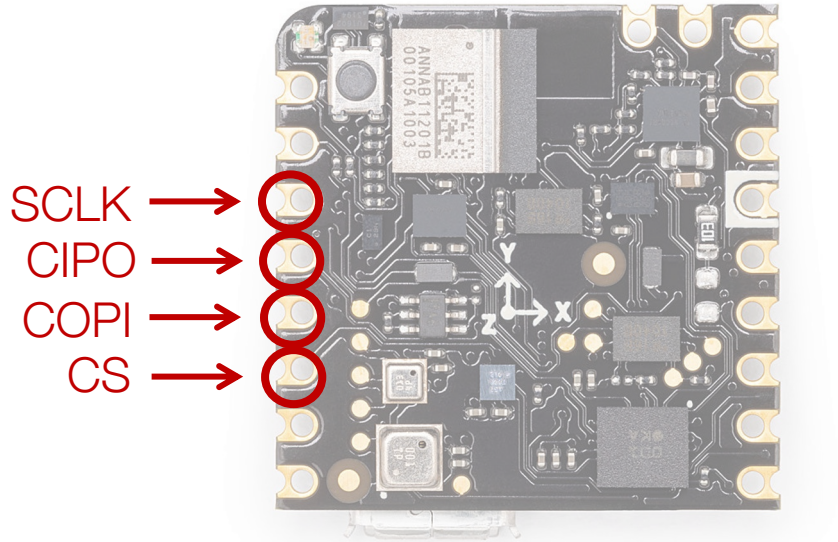
- MOSI (Master Out Slave In): master sends data
- MISO (Master In Slave Out): slave sends data
- SCLK (Serial Clock): generated by the master
- SS/CS (Slave Select / Chip Select): line dedicated to slave selection. We need a CS dedicated to each connected slave device



Serial Peripheral Interface (SPI)

```
#include <SPI.h>

const int chipSelectPin = 10; // Define a pin CS. It allows to activate the external device
```

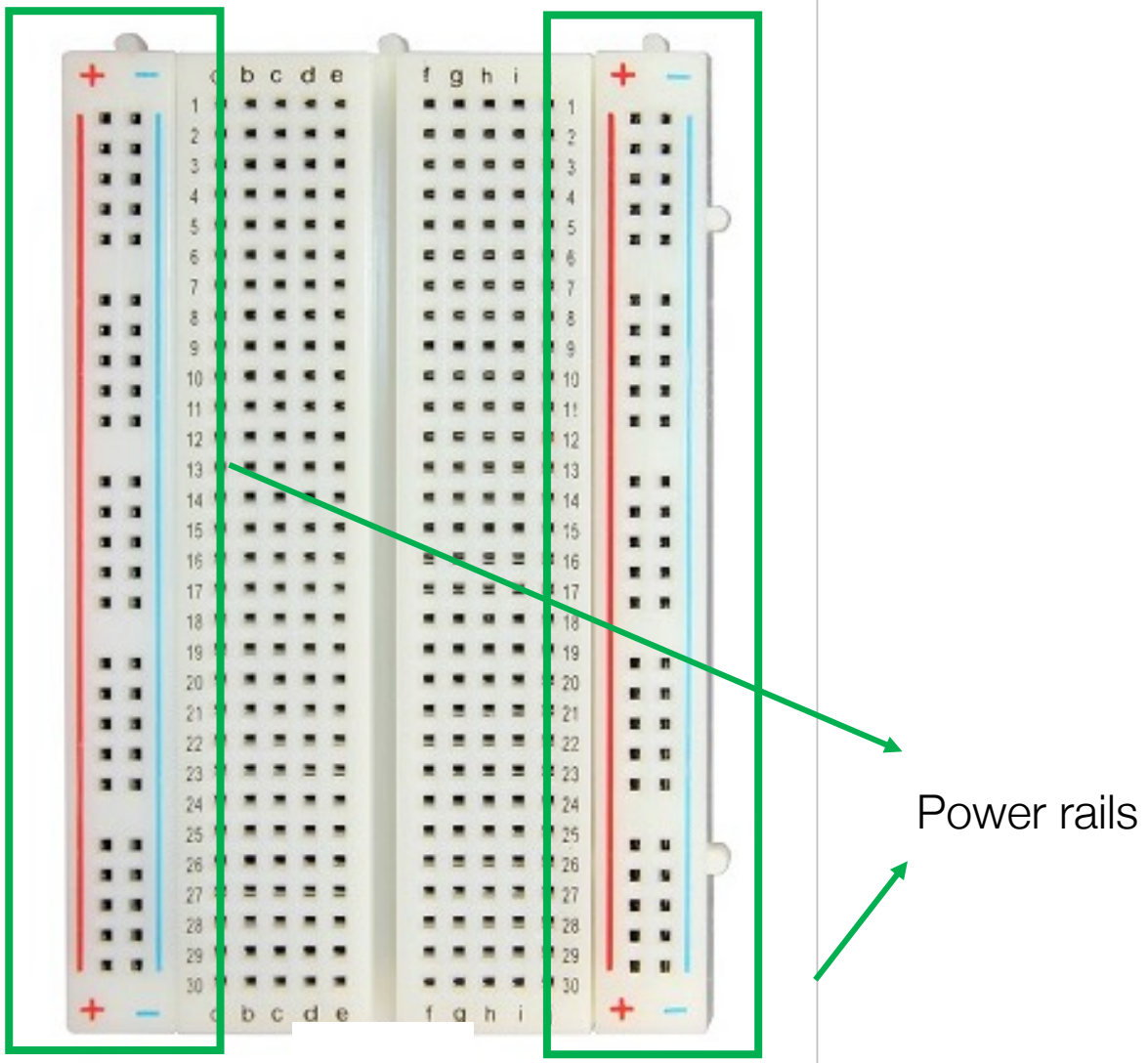
```
void setup() {
    pinMode(chipSelectPin, OUTPUT); // Set the chip select pin as output
    digitalWrite(chipSelectPin, HIGH); // Pull the CS pin HIGH to unselect the device
    SPI.begin(); // Initialize the SPI communication
}
```

```
void loop(){
    byte address = 0x35; // Address or command
    byte value = 0xFA; // Value to send
    digitalWrite(chipSelectPin, LOW); // Pull the CS pin LOW to select the connected device
    SPI.transfer(address); // Send the address
    SPI.transfer(value); // Send the value
    digitalWrite(chipSelectPin, HIGH); // Pull the CS pin HIGH to unselect the device
}
```

How to read data from sensors?

Measurements with external sensors

A **breadboard** is a tool used for prototyping and testing electronic circuits without soldering



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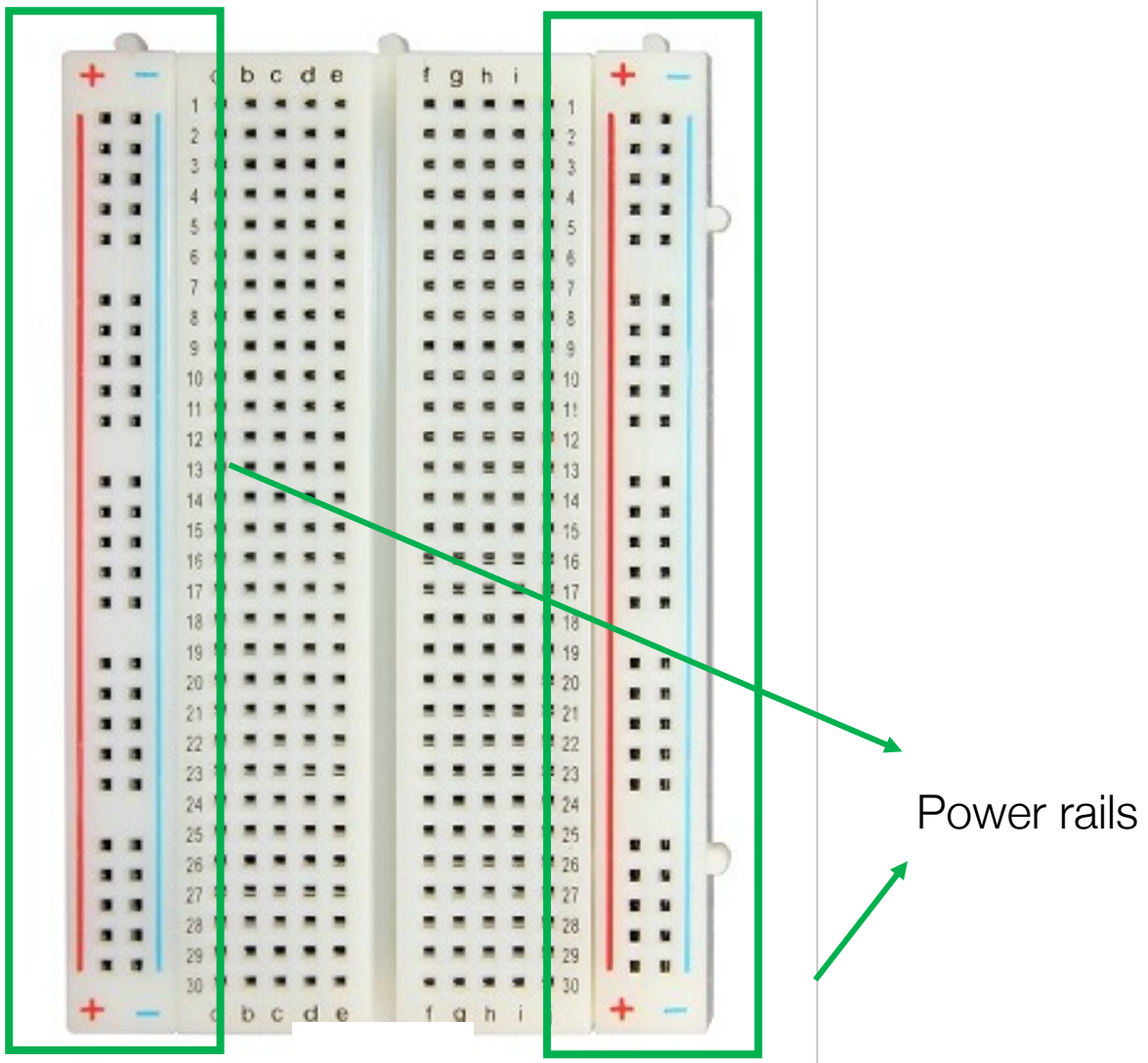
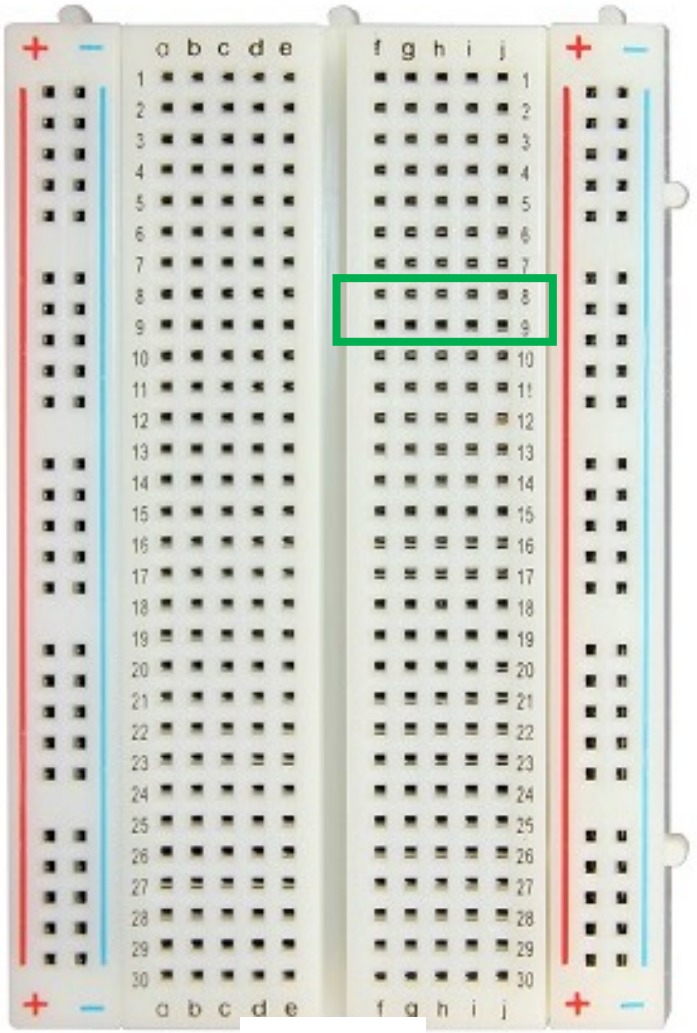


Measurements with external sensors

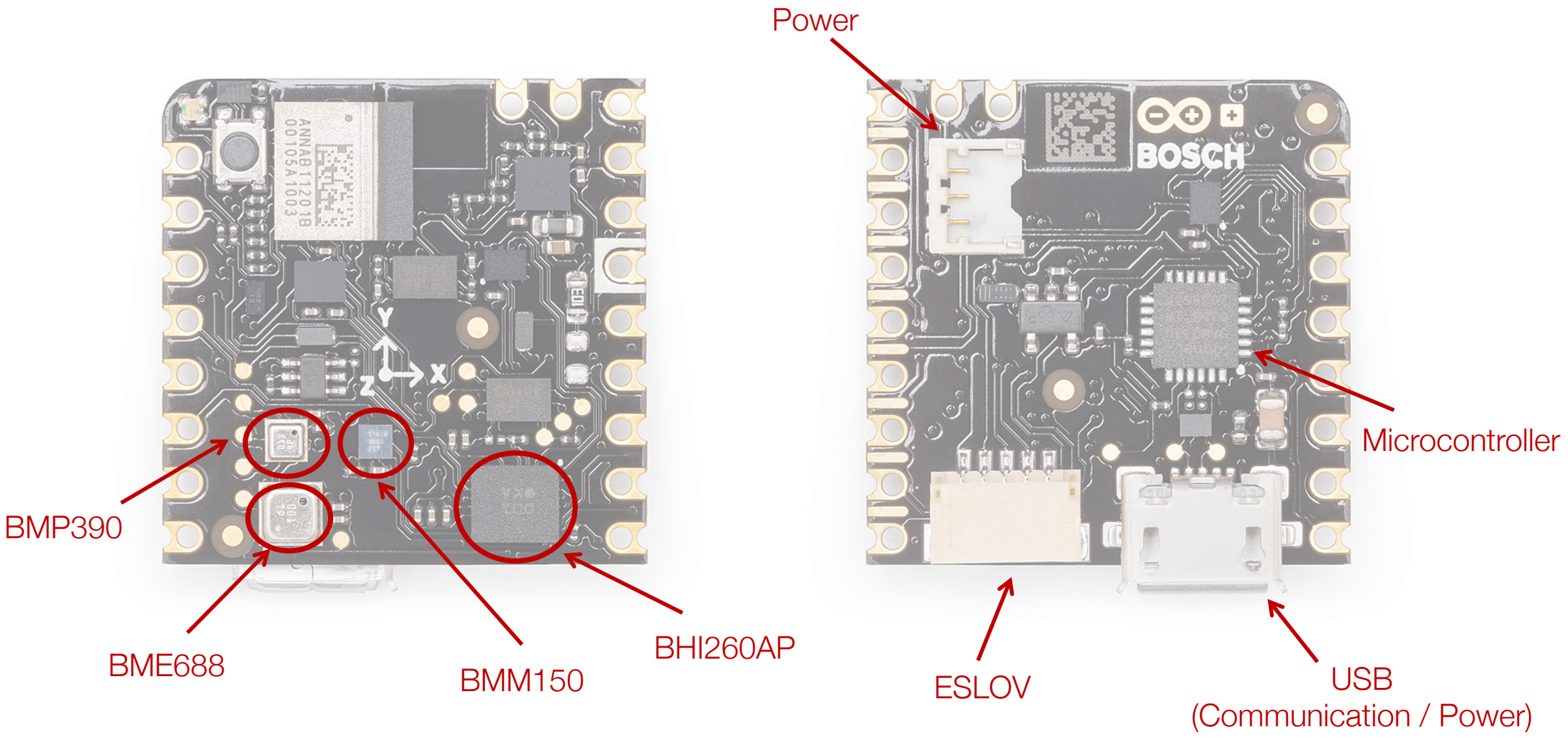
Terminal Strips: The middle area of the breadboard is divided into rows and columns.



The rows are typically marked by numbers (e.g., 1, 2, 3, etc.). Each row contains a set of connected terminals that are electrically connected in parallel..

The columns are typically marked by letters (e.g., A, B, C, D, E). Components placed in the same column within the same row are not electrically connected.

Nicla Sense ME



Embedded sensors

There are three ways to read from the on-board sensors:

1. Read the sensors directly from Nicla Sense ME in standalone mode.
2. Read sensor values through I2C by connecting an ESLOV cable.
3. Read sensor values through Bluetooth Low Energy.

https://docs.arduino.cc/tutorials/nicla-sense-me/cheat-sheet/

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In the **standalone mode**, the sensors can be accessed through **sensor objects**. The sensor data is then read by Nicla Sense ME's on-board microcontroller.

Libraries that we need to install to read from the sensors in any of these mode:

- **Arduino_BHY2**

To install libraries: from the Arduino IDE, select Tools->Manage Libraries

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Embedded sensors

To use the sensors in your sketches, you need to know the sensors ID, which can be found in:
https://docs.arduino.cc/tutorials/nicla-sense-me/cheat-sheet/
https://github.com/arduino-libraries/Arduino_BHY2/blob/main/src/sensors/SensorID.h

Sensors can be divided in classes:

- **Sensor**: all the other sensors which have a single value to be read (value property)
- **SensorOrientation**: sensors with the Euler format, allows to read pitch, roll, heading property
- **SensorXYZ**: sensors with XYZ format
- **SensorQuaternion**: sensor with the quaternion format (rotation, game rotation and geomagnetic rotation vector)
- **SensorActivity**: sensor with the activity format (getActivity function)
- **SensorBSEC**: Bosch Sensortec Environmental Cluster (IAQ data)

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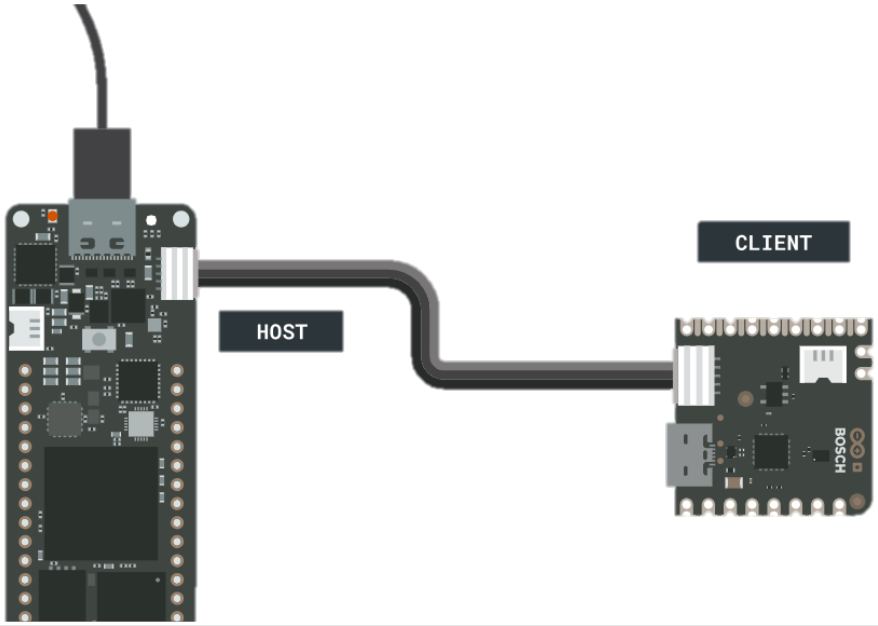
The MKR WiFi 1010 (host) communicates with Nicla Sense ME via the **BHY2Host library.**

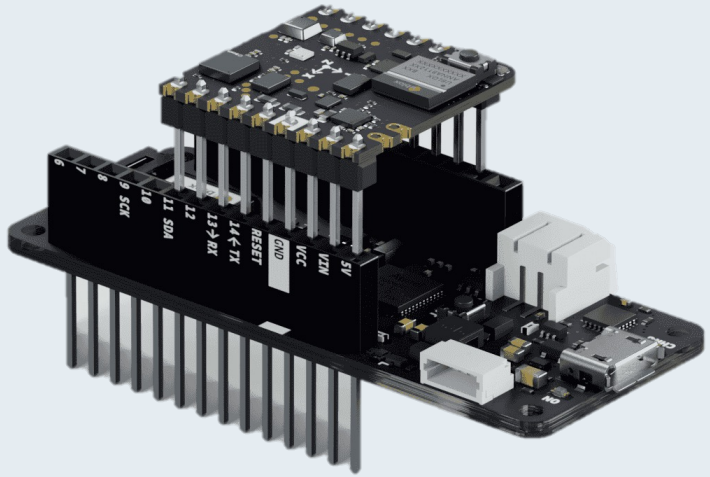


https://docs.arduino.cc/tutorials/nicla-sense-me/use-as-mkr-shield/

Arduino Nicla Sense ME as a MKR Shield

1. Solder two rows of headers: position the long side of the headers on the **battery connectors side.**

2. To connect the boards, plug the Nicla Sense ME on top of the MKR board, ensuring the **LED and Reset button** on the Nicla are facing upwards.



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